tutorial

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We will explain the procedure for developing a program for WonderWitch step by step from setup to execution.
We will also introduce the sample programs included in the kit.

table of contents

- First thing to know
- setup
- Editing source files
- compile
- Convert to WonderWitch transfer format
- send on
- execution
- Sample list
- Implementation and use of user IL

First thing to know

Development procedure

Follow the steps below to develop a program that runs on a WonderWitch cartridge.

1. Development environment construction (setup)
2. Editing source files
3. Compiling (or assembling) source files
4. Object file link
5. Conversion to formats available on WonderWitch

File

Various types of files are used when developing programs. Here, we will briefly explain the contents and roles of the files, which are typical ones.

file name	Contents
* .c	C source file. Describe the main part of the program.
* .h	C header file. Define constants used in .c files and declare functions / types.
* .obj	Object file. Compile, assemble and generate the source files.
* .lib	Library file. Generate object files together with a librarian.
* .bin	Linked binary file. Generate by linking .obj file and .lib file.
* .fx	WonderWitch transfer format file. The file is converted to a format that can be placed on WonderWitch with the mkfent command . The information required for the conversion is specified in the .cf file.
* .cf	mkfent information file. Describes the information used when converting a file to WonderWitch transfer format with the mkfent command. For the detailed format, refer to "How to use mkfent" in the reference manual .

makefile	Dependency description file for make command. Use the make command and kmmake command to describe the dependencies between files and the command to be executed when a series of steps are automatically executed.

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setup

To develop a program using WonderWitch, you need to install the necessary tools on your PC and make some settings.
Please read the following instructions carefully and set them appropriately for your computer environment.

* Information on settings in various environments other than Windows (outside the guaranteed operation range) is available on the WonderWitch support site.

Installation of tools

Install the tools from the Magical CD-ROM to your PC according to "Installing to PC" in the user's manual .

Edit configuration file

Edit the contents of the configuration files under the WWitch folder as appropriate according to the actual installation location.

The standard installation location and the description to be corrected are as follows.

Installation contents	Standard installation location	Modification target
WWitch folder	`C:¥WWitch`	`C:¥WWitch` All the parts of the following files • bin \ setup.bat • lsic86 \ bin \ _lcc86ww
Turbo C (only when using)	`C:¥TC`	`C:¥TC` All the parts of the following files • bin \ setup.bat

Launch and configure the MS-DOS prompt

Program development is done using the MS-DOS prompt. For Windows98, start Menu → Programs → MS-DOS Prompt. For Windows 2000, use Start Menu → Programs → Accessories → Command Prompt, and for WindowsNT 4.x, use Start Menu → Programs → Command Prompt.

To use the development tools correctly on the MS-DOS prompt, you need to set the command search path and so on. A batch file (setup.bat) for making this setting is prepared in the WWitch \ bin folder, so execute it.

```
> \ WWitch \ bin \ setup.bat
```

* Specify the path according to the actual installation location.

Editing source files

Create a source file using a text editor.

For example, if the program displays "Hello, World!" In the upper left of the screen, write as follows.

```
#include <sys / bios.h>

void main ()
```

```
{
    text_screen_init ();
    text_put_string (0, 0, "Hello, World!");
}
```

When you have finished editing the program .c, save it to a file with a name ending in.
At this time, give a name that appropriately describes the contents of the program. For example, for the above program, a name like hello.c would be good.

The text editor used for editing may be the one you normally use, such as Notepad or Hidemaru, but please note the following points.

- Save in text format
- The extension is .c

Especially if you are using Notepad, please note that the file is saved with the extension .txt by default.

* To be able to compile with either LSI C-86 for WonderWitch or Turbo C 2.0, name the source file so that it fits in the file name of 8 characters + extension of 3 characters.

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compile

Process the source file using the compiler to get the executable format.

The included compiler runs on the MS-DOS prompt. For example, when compiling the above hello.c using the included LSI C-86 for WonderWitch, use the MS-DOS prompt (→ MS-DOS prompt startup and settings) that has been initially set up. , Do the following:

```
> lcc86 -o hello.bin hello.c
```

If there is no problem with the contents of the program and the compilation is successful, a file called hello.bin will be generated.
If something goes wrong, the compiler will display an error message. In this case, the hello.bin file will not be generated.
For example, if you change the place that should be; (semicolon) to, (comma), you will get the following error message.

```
hello.c 7: syntax error near'}'
```

The error message indicates the file name, the line number where the error was detected, and the content of the error, so modify and compile the source file repeatedly until the error message disappears.

Convert to WonderWitch transfer format

After generating the .bin file, convert it to a format that can be transferred to the WonderWitch dedicated cartridge (WonderWitch transfer format).
To convert on a Windows OS, use the included file format conversion tool "mkfent".

mkfent converts the .bin file to a .fx file according to what you wrote in the .cf file.

For example, the hello.bin file

- Placed on WonderWitch with the name hello
- The description of the file is "Hello"
- Read / write, executable
- The output file name on the PC is hello.fx

To convert under the condition

```
name: hello
info: Hello
mode: 7
source: hello.bin
output: hello.fx
```

Prepare a file (let's call it hello.cf) described as, and execute the mkfent command.

```
mkfent hello.cf
```

This will generate hello.fx.

For the detailed format of the .cf file, refer to "How to use mkfent" .

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send on

Transfer the .fx file generated by the mkfent command to the WonderWitch dedicated cartridge.

To transfer files on a Windows OS, use the included file transfer tool "TransMagic".

For details on how to use TransMagic, refer to "Transfer to Cartridge" in the user's manual.

For non-Windows systems, use communication software that supports the XMODEM protocol. For details, refer to "Advanced Usage" in the User's Manual .

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execution

The program transferred to the WonderWitch dedicated cartridge is executed using the shell function.

For details on how to execute with the default shell "Meg", refer to "Running a program and manipulating files" in the user's manual .

Using the make command

The above is the flow of program development in WonderWitch. It requires a great many steps and may have to be repeated over and over again.

If you write the dependencies between files in a file named makefile, you can automatically execute a series of steps just by executing the make command without having to execute it manually every time. It is very convenient when you make many corrections and compilations, so we recommend you to use it.

The LSI C-86 for WonderWitch folder contains the kmmake command, and the Turbo C 2.0 folder contains the make command. Please refer to the manual of each command and use it.

For how to write makefile, the makefile of each sample stored in the samples folder will be helpful. Copy makefile.lcc from the samples folder to makefile.inc to use LSI C-86 for WonderWitch as the default compiler, and copy makefile.tcc to makefile.inc to use Turbo C 2.0.

The makefile in each sample folder follows the format of the kmmake command. Please note that if you use the make command that comes with Turbo C, you will need to make corrections.

Sample list

The WWitch \ samples folder contains samples that show how to use WonderWitch's features. Not all feature samples are available, but they are always useful when programming in WonderWitch, so please refer to them.

hello hello
: Display a message on the screen.
  Text screen BIOS and key BIOS samples.
  It is also a sample that references the command line arguments passed in the remote shell.

calendar

Display the calendar.
This is a sample of the RTC function.
The year and month change with the X key group.

random
: Display random numbers.
This is a sample of the standard library function.

scroll
: Scroll the screen.
This is a sample screen control BIOS.
Scroll up / down / left / right with the X key group.

copy
: Copy the contents of the file.
This is a sample file access using the FILE structure.
Execute it in a remote shell, and specify the file names of the copy source and copy destination with command line arguments.

ls
: Display a list of files stored in / rom0.
A sample low-level I / O library function.

version
: Displays the version of FreyaOS and FreyaBIOS.
This is a sample of the standard library function.

pid
: Displays information about the process.
This is a sample of the process status acquisition function group.

ps
: Lists information about processes.
This is a sample of the process status acquisition function group.

yield
: Stops the execution of a process and delegates the execution right to another process.
This is a sample of the functions of ProcIL.
The calendar also has a yield function, and if you execute it while switching to this sample, you can experience the process switching.

il
: Use the indirect library to display a message on the screen.
This is a sample of the indirect library. Since footest.fx (on WonderWitch, footest) calls the function of foo.il (same @foo), please transfer foo.il in addition to footest.fx.

procil
: Using the user-defined ProcIL, a message is displayed on the communication line each time the process control function group is called.
This is a sample that replaces the standard indirect library. Transfer proc.il (@proc on WonderWitch), connect to communication software such as TeraTerm, and then restart the OS.
